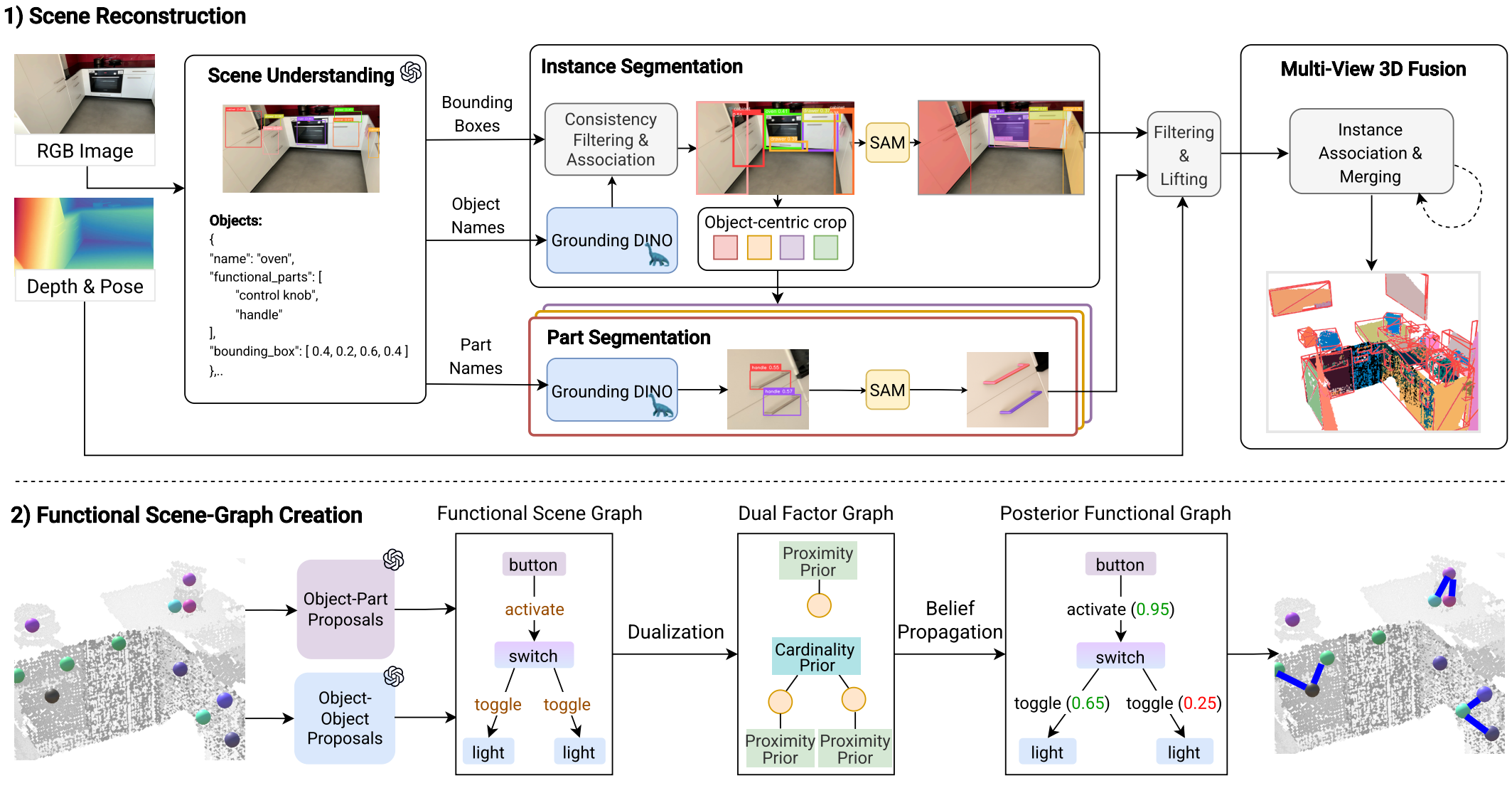
1) Scene Reconstruction
RGB Image
Depth & Pose
Scene Understanding
Objects:
{
"name": "oven",
"functional_parts": [
"control knob",
"handle"
],
"bounding_box": [0.4, 0.2, 0.6, 0.4]
},..
Bounding Boxes
Object Names
Part Names
Instance Segmentation
Consistency Filtering & Association
SAM
Grounding DINO
Object-centric crop
Part Segmentation
Grounding DINO
SAM
Filtering & Lifting
Multi-View 3D Fusion
Instance Association & Merging
2) Functional Scene-Graph Creation
Object-Part Proposals
Object-Object Proposals
Functional Scene Graph
button
activate
switch
toggle
toggle
light
light
Dualization
Dual Factor Graph
Proximity Prior
Cardinality Prior
Proximity Prior
Proximity Prior
Belief Propagation
Posterior Functional Graph
button
activate (0.95)
switch
toggle (0.65)
toggle (0.25)
light
light